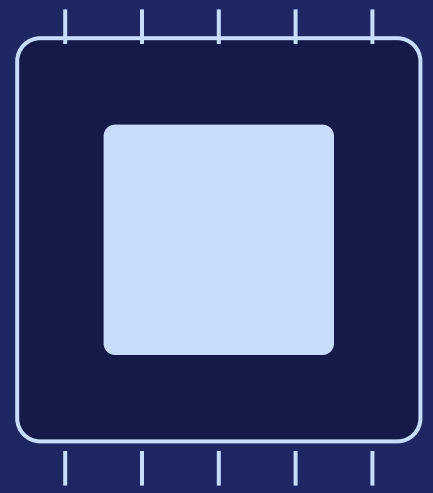
CPE17 · MODULE 1

Computer Architecture & Organization

History, Overview of Computer Architecture, and Relevant Tools, Standards, and Engineering Constraints



Dalubhasaan ng Lungsod ng San Pablo· First Semester, 2024–2025

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Module Overview

Self-paced learning module details

COURSE	SEMESTER / AY	MODULE NO.	WEEK DURATION	DATE
CPE17	First Semester, 2024–2025	1	Week 2–3	August 26 – September 7, 2024

LESSON TITLE

History, Overview of Computer Architecture and Relevant Tools, Standards and/or Engineering Constraints

DESCRIPTION OF THE LESSON

In this lesson, students will learn concepts about Computer Architecture and Organization — including the parts and components of a computer system, a brief history of computers, and the relevant tools and constraints in engineering design.

Learning Outcomes

What students should walk away with

INTENDED LEARNING OUTCOMES

- Learn about the concepts of Computer Architecture and Organization
- Learn the parts and components of a computer system, and a brief history of computers
- Learn the relevant tools and constraints in engineering design

TARGETS / OBJECTIVES

- Familiarize themselves with the concepts of Computer Architecture and Organization
- Know the parts and components of a computer system, and a brief history of computers
- Identify the relevant tools and constraints in engineering design

Two Sides of the Same Machine

Architecture is the “what” — Organization is the “how”

COMPUTER ARCHITECTURE

The conceptual design and fundamental operational structure of a computer system. It deals with the high-level design of a computer’s hardware components and their interactions.

Includes: instruction set, data types, addressing modes, and visible registers.

COMPUTER ORGANIZATION

The operational structure and implementation of computer systems. It focuses on how the hardware components are connected and interact with each other to execute instructions.

Includes: how circuits, signals, and physical hardware are actually wired together.

Architecture VS Organization

A useful way to remember the distinction:

	Computer Architecture	Computer Organization
Question it answers	*What* can the system do?	*How* does it do it?
Audience	Programmers, compiler writers	Hardware/circuit engineers
Examples	Instruction set, addressing modes, visible registers	Datapaths, control signals, pipelining, bus wiring
Analogy	The car's dashboard and controls	The car's engine and transmission

Key Aspects

Three concepts that tie architecture and organization together



Instruction Set Architecture

Defines the set of instructions the CPU can execute and how they are encoded.



Microarchitecture

The implementation of the ISA — including CPU design, pipelining, and parallel processing.



Memory Hierarchy

The organization of different memory types (registers, cache, main, secondary) to optimize access time and performance.

Computer Design: Basic Components

Planning how the different parts of a computer work together

1	**CPU**	Central Processing Unit — executes instructions and performs calculations
2	**Memory**	Includes RAM (Random Access Memory) — holds running programs and data
3	**Storage**	Retains data permanently, even when the system is powered off
4	**I/O Systems**	Input/Output — connects the system to the outside world
5	**Bus Systems**	Pathways that carry address, data, and control signals between components

Brief Computer Architecture History

Six generations of computing

0	1	2	3	4	5
Mechanical Computers	**Vacuum Tubes**	**Transistors**	**Integrated Circuits**	**Microprocessors**	**Artificial Intelligence**
1642–1945	1945–1955	1955–1965	1965–1980	1980–?	present

History Highlights

Milestones that shaped modern computing

1642

Pascal's Calculator

One of the first mechanical calculators, by Blaise Pascal.

1837

Babbage's Analytical Engine

Never completed, but laid the groundwork for modern computing.

1945

ENIAC

First general-purpose electronic digital computer, using vacuum tubes.

1945

Stored-Program Concept

John von Neumann's idea: instructions stored in memory alongside data.

1964

IBM System/360

A family of compatible computers using integrated circuits.

1971

Intel 4004

First commercially available microprocessor, enabling personal computers.

1980s+

Personal Computers

Widespread PC adoption brought GUIs, the internet, and modern software.

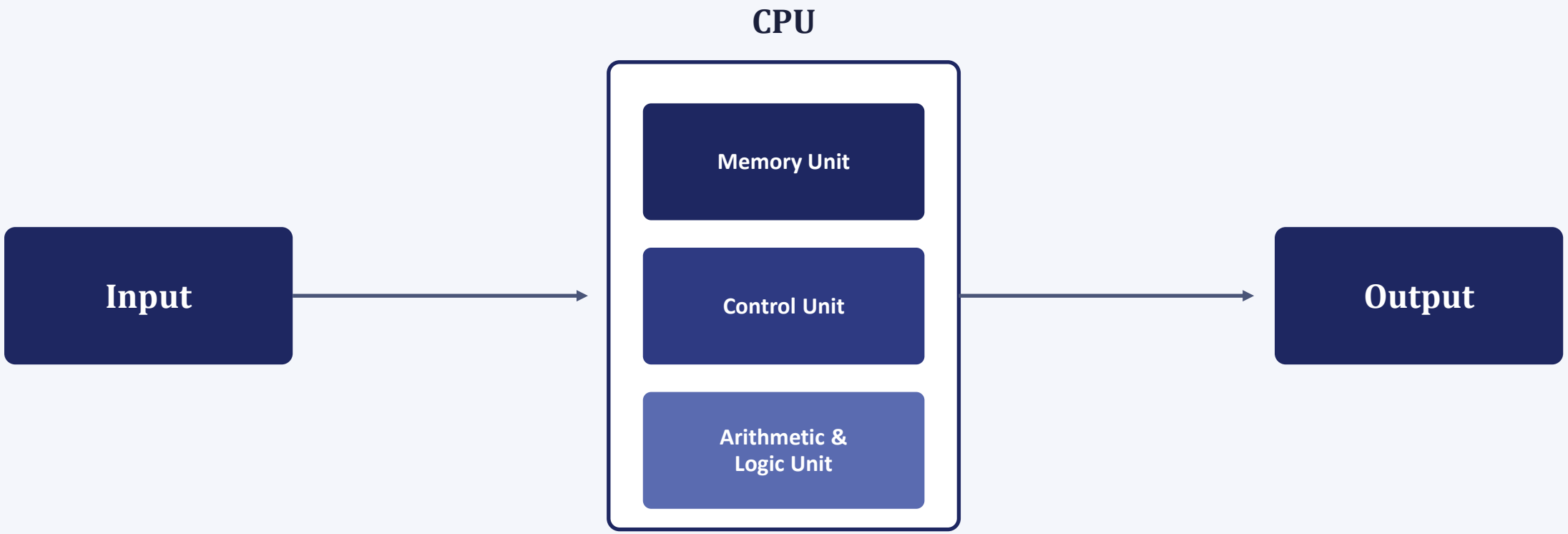
Today

AI & Quantum Computing

Self-learning systems and quantum/neuromorphic computing push the frontier.

Computer Systems Organization

A network of processors, memory, and I/O devices working together



CPU

Fetches program and data from memory, performs arithmetic/logical operations, stores results back.

ALU

Performs all arithmetic and logic operations required by an instruction.

CU

Controls sequential instruction execution and guides data flow through the system.

Input / Output

Input devices convert data to digital form; output devices convert digital data to human-readable form.

Inside the CPU: Processors

The “brain” of the computer — fetches, examines, and executes

CPU Organization	The datapath consists of registers, the ALU, and buses connecting the pieces; registers feed two ALU input registers.
Instruction Execution	Each instruction runs through the fetch–decode–execute cycle — central to how every computer operates.
RISC vs. CISC	RISC (Reduced Instruction Set Computer) is contrasted with CISC (Complex Instruction Set Computer).
Instruction-Level Parallelism	The parallel or simultaneous execution of a sequence of instructions in a program.
Processor-Level Parallelism	In a multiprocessor system, each processor executes a different thread on the same or different data.

Primary Memory

Where programs and data are stored during execution

Bits

The basic unit of memory — a binary digit that holds a 0 or a 1.

Memory Address

Each memory cell has a unique number, its address, by which programs refer to it.

Byte Ordering

Bytes in a word can be numbered left-to-right or right-to-left — a choice with major implications.

Error-Correcting Codes

Guards against occasional errors using error-detecting or error-correcting codes.

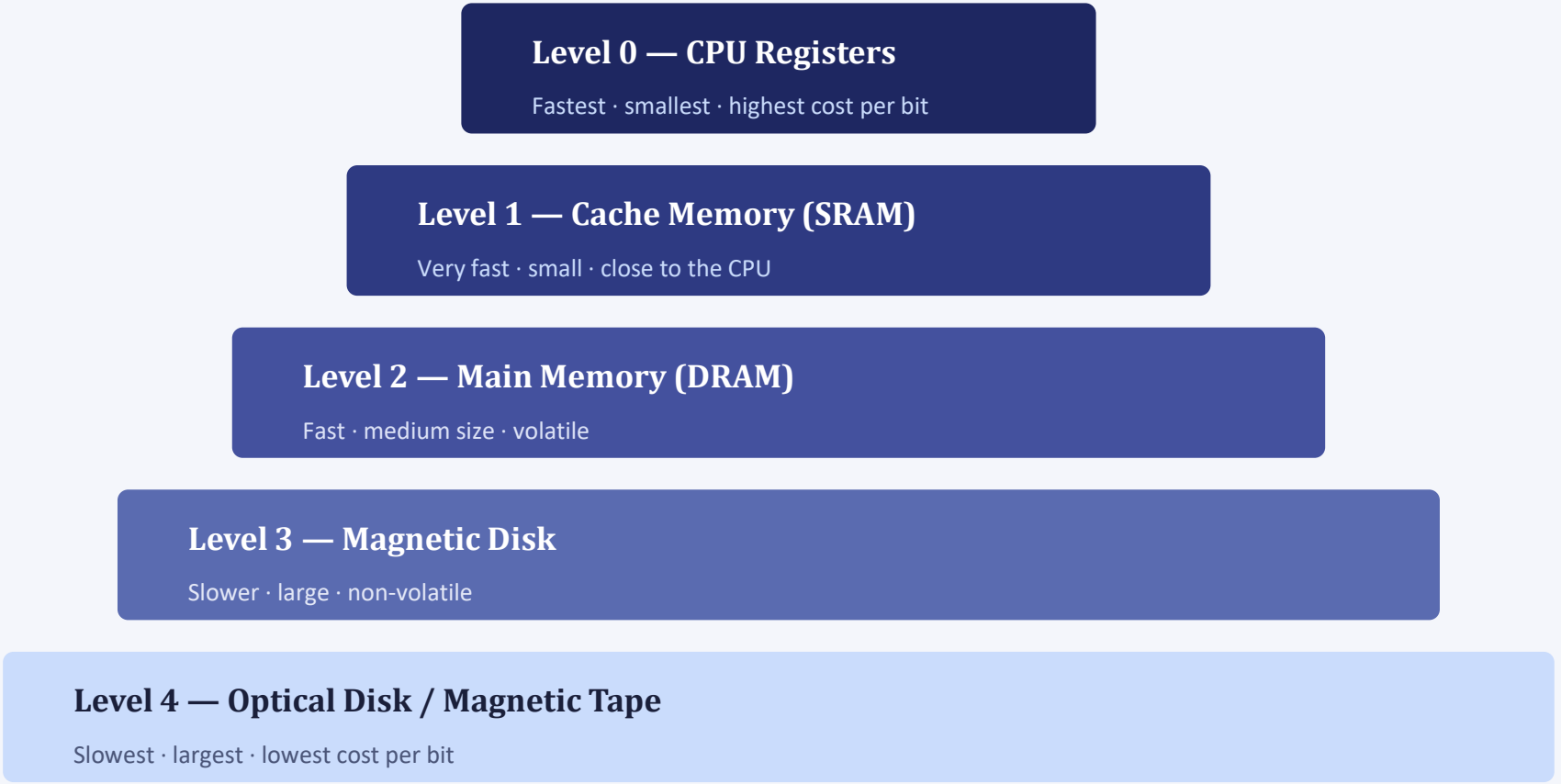
Cache Memory

Temporarily stores frequently used instructions and data for faster CPU access.

The Memory Hierarchy

Organizing memory to minimize access time (locality of reference)

↑ SPEED & COST



↓ CAPACITY

Secondary Memory & Storage Types

Main memory is always too small — secondary storage fills the gap

1

Magnetic Storage

Magnetic disk · Floppy disk · IDE · SCSI · RAID

2

Optical Storage

CD-ROM · DVD · Blu-Ray

3

Flash Storage

USB flash drive · HDD · SSD

4

Cloud Storage

Dropbox · iCloud · Google Drive · OneDrive · Mega · Box

Input / Output & Buses

A bus carries address, data, and control signals between components



I/O Bus — interfaces translate signals between the CPU/memory bus and each device.

Monitor **Keyboard** **Hard-disk**

Terminals: Keyboard + monitor for entering and transcribing data

Mouse: Hand-controlled device for GUI interaction

Printers: Produces paper copies of documents or data

Digital Camera: Captures photographs and stores them as data

Headset: Connects to talk/listen hands-free

Character Code: Unique numbers assigned to characters (e.g. ASCII)

Relevant Tools & Standards

Four components required to implement a computerized input-process-output model

Computer Hardware

The physical mechanisms to input, output, manipulate, and process data — the tangible part of the system.

Computer Software

Programs that tell the computer what to do: system software (e.g. the OS) and application software (e.g. Word, Firefox).

Data

The information being manipulated — numeric, alphanumeric, graphic, or another form the computer can represent.

Communication Component

Hardware and software that transport programs and data between interconnected computer systems.

Architecture Is Engineering Design Under Constraints

Eight factors that shape every design decision

1

Functionality

Does the design work and meet user needs? (usability, satisfaction, efficiency, effectiveness)

2

Cost

Resources needed to conceive, design, produce, and maintain the product

3

Performance

Coordination of conditions that produce desired results: input, output, efficiency, quality, outcome

4

Size

Physical space taken up by the system solution

5

Power & Energy

Energy required to perform its function — often traded off against cost

6

Time to Market

Time from concept to release — or until the first unit is sold

7

Maintainability

Probability a failed component can be restored within a specified time

8

Safety

Embedding health & safety considerations from the earliest design stages

Key Takeaways

1. Architecture defines what a computer can do; organization defines how it does it.
2. A computer system is built from the CPU, memory, storage, I/O, and a shared bus.
3. Computing has evolved through six generations — from mechanical devices to AI.
4. The CPU cycles through fetch–decode–execute for every single instruction.
5. Memory is organized in a hierarchy that trades speed for capacity.
6. Good architecture is engineering design under real-world constraints: functionality, cost, performance, size, power, time-to-market, maintainability, and safety.